RECONNAISSANCE & ENUMERATION

1. Objective

The objective of this task was to understand and map an application's attack surface using reconnaissance and enumeration techniques.

2. Concept/Understanding

Reconnaissance is the first phase of penetration testing.

Two types:

- **Passive Recon:** No direct interaction, including WHOIS, DNS, and certificates.
- **Active Recon:** Direct interaction, including port scanning and brute forcing.

Key concepts:

- Technology identification
- Subdomain discovery
- Directory and endpoint discovery
- Parameter discovery
- Login page identification
- API endpoint discovery
- Hidden functionality discovery

3. Practical Work

Completed three TryHackMe rooms:

- Passive Recon
- Active Recon
- Content Discovery

Used tools:

- WHOIS for domain information
- Nslookup for DNS records
- Nmap for port scanning
- Gobuster for directory brute forcing
- Browser Developer Tools for analysis

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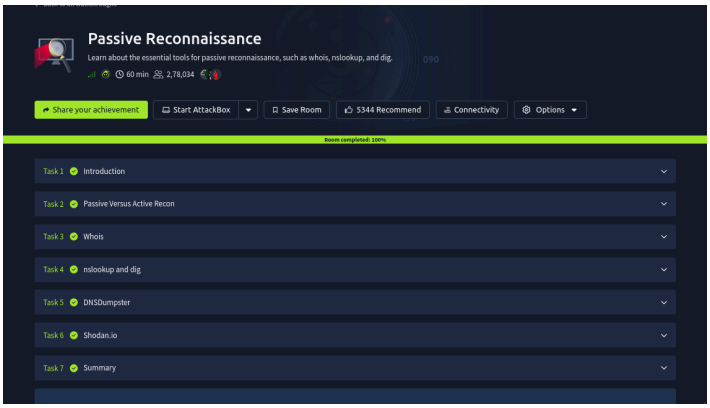
4. Tools Used

- TryHackMe
- WHOIS
- Nslookup / Dig
- Nmap
- Gobuster
- Browser Developer Tools

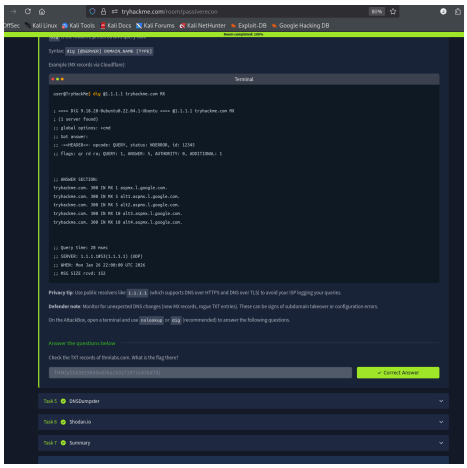
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5. Evidence

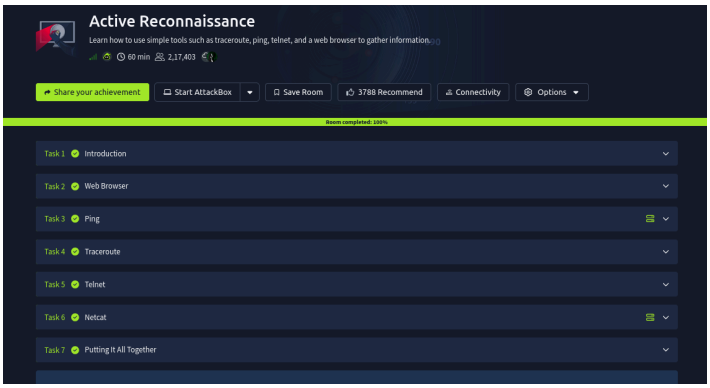
Passive Recon completion



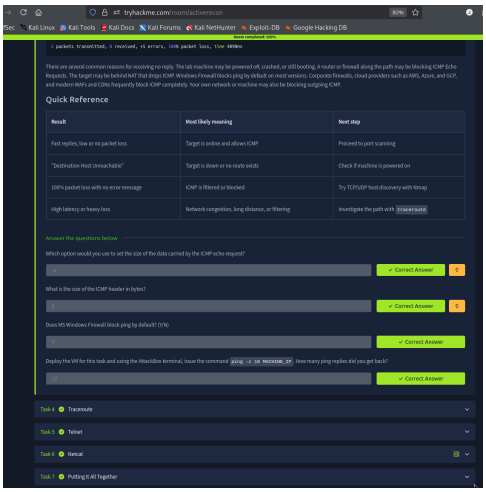
Passive Recon answers



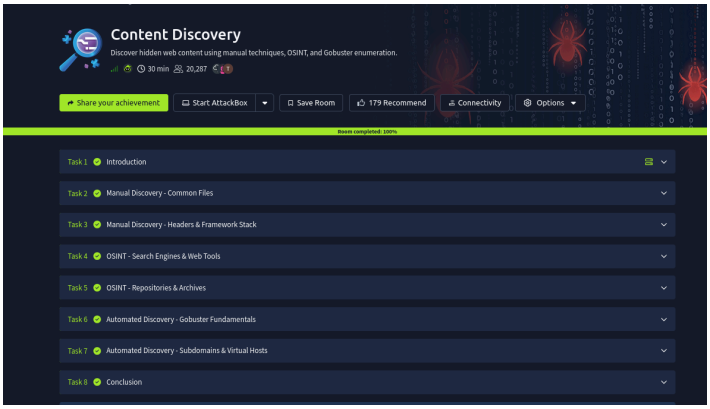
Active Recon completion



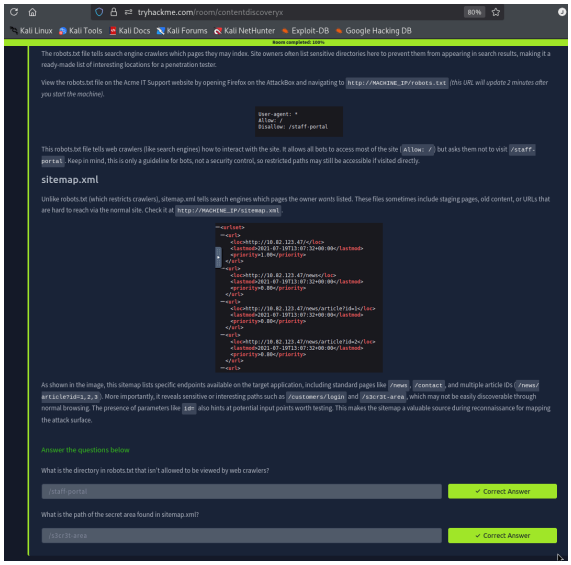
Active Recon answers



Content Discovery completion



Content Discovery answers



WHOIS output



```
$ whois tryhackme.com
   Domain Name: TRYHACKME.COM
   Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.namecheap.com
   Registrar URL: http://www.namecheap.com
   Updated Date: 2025-05-11T14:06:02Z
   Creation Date: 2018-07-05T19:46:15Z
   Registry Expiry Date: 2034-07-05T19:46:15Z
   Registrar: NameCheap, Inc.
   Registrar IANA ID: 1068
   Registrar Abuse Contact Email: abuse@namecheap.com
   Registrar Abuse Contact Phone: +1.6613102107
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Name Server: KIP.NS.CLOUDFLARE.COM
   Name Server: UMA.NS.CLOUDFLARE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-10-12T03:12:24Z <<<

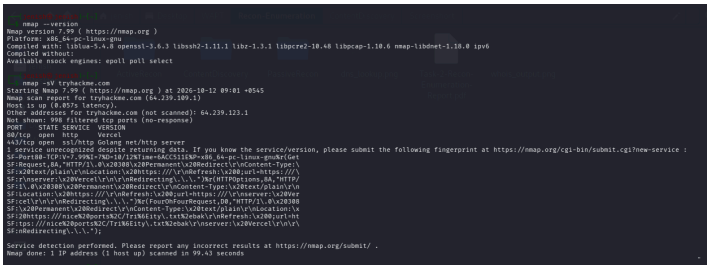
For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

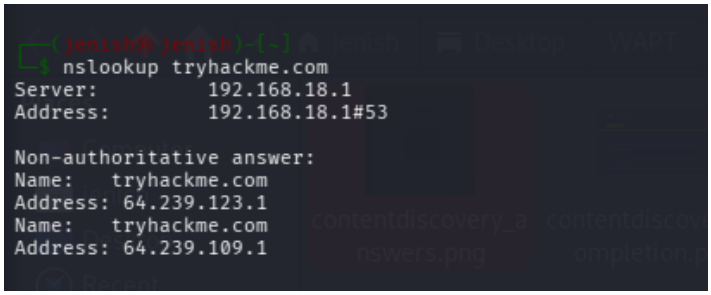
TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability.  VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain name: tryhackme.com
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2025-05-11T14:06:03.00Z
Creation Date: 2018-07-05T19:46:15.00Z
Registrar Registration Expiration Date: 2034-07-05T19:46:15.00Z
Registrar: NAMECHEAP INC
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.9854014545
Reseller: NAMECHEAP INC
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Registry Registrant ID:
Registrant Name: Redacted for Privacy Purposes
Registrant Organization: Privacy service provided by WITHHELD FOR PRIVACY LLC
Registrant Street: 16192 Coastal Highway
Registrant City: Lewes
Registrant State/Province: Delaware
Registrant Postal Code: 19958
Registrant Country: US
Registrant Phone: +1.3022061391
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: a70a4ff6d25041a4837899719 4f9e834.protect@withheldforprivacy.com
Registry Admin ID:
Admin Name: Redacted for Privacy Purposes
Admin Organization: Privacy service provided by WITHHELD FOR PRIVACY LLC
Admin Street: 16192 Coastal Highway
Admin City: Lewes
Admin State/Province: Delaware
Admin Postal Code: 19958
Admin Country: US
Admin Phone: +1.3022061391
Admin Phone Ext:
Admin Fax:
Admin Fax Ext:
Admin Email: a70a4ff6d25041a48378997194f9e834.protect@withheldforprivacy.com
Registry Tech ID:
Tech Name: Redacted for Privacy Purposes
Tech Organization: Privacy service provided by WITHHELD FOR PRIVACY LLC
Tech Street: 16192 Coastal Highway
Tech City: Lewes
Tech State/Province: Delaware
Tech Postal Code: 19958
```

Nmap scan output



DNS lookup output



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6. Result

Successfully:

- Understood passive vs. active recon
- Gathered domain information
- Discovered subdomains
- Learned directory brute forcing
- Identified technologies
- Mapped the attack surface

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7. Key Learnings

- Recon is the foundation of pentesting
- Passive recon leaves no traces
- Active recon is thorough but detectable
- Enumeration reveals hidden attack surface
- More information can make an attack easier
- Tools such as Nmap, Gobuster, and WHOIS are essential

Conclusion

This work established a foundational understanding of reconnaissance and enumeration and their role in mapping an application's attack surface.
